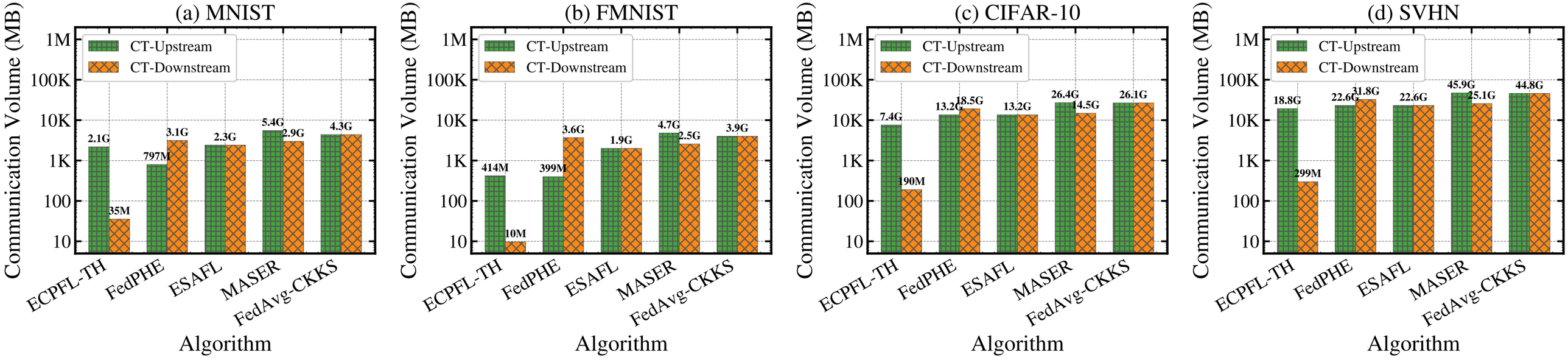
(a) MNIST
(b) FMNIST
(c) CIFAR-10
(d) SVHN
Communication Volume (MB)
1M
100K
10K
1K
100
10
CT-Upstream
CT-Downstream
2.1G
35M
797M
3.1G
2.3G
5.4G
2.9G
4.3G
414M
10M
399M
3.6G
1.9G
4.7G
2.5G
3.9G
7.4G
190M
13.2G
18.5G
13.2G
26.4G
14.5G
26.1G
18.8G
299M
22.6G
31.8G
22.6G
45.9G
25.1G
44.8G
ECPFL-TH
FedPHE
ESAFL
MASER
FedAvg-CKKS
Algorithm